

❖ Final FMT (35)

Forensic

1) About nuclear DNA which statement is false:

- a) it is valid for long time
- b) it has short validity only
- c) it resist putrefaction for long time
- d) A&C

2) Sure ocular sign for death:

- a) sub conjunctival hemorrhage
- b) constricted pupil
- c) trucking
- d) congestion

3) All cause physical block in airway except:

- a) Suffocation
- b) Choking
- c) Postural asphyxia
- d) Drowning

4) Presence of weeds and froth in airway passage indicate:

- a) Salt water
- b) Fresh water
- c) Postmortem drowning
- d) Antemortem drowning

5) All are true about brain concussion except:

- a) momentary arrest or loss of all functions of the brain
- b) can result from positive and negative wave of an explosion
- c) not a pathological condition
- d) may lead to brain compression

6) Natural cause of death in infant:

- a) difficult labor
- b) head injury
- c) drowning
- d) choking

7) Character of exit wound:

- a) Blacking
- b) Irregular
- c) Small
- d) Internal beveling

8) Leucocrystal violet is preliminary test used for latent blood stain:

- a) True
- b) False

9) All are characters of respired lung except:

- a) Solid consistency
- b) Rounded edge
- c) Mottled appearance
- d) high weight

10) Hydrostatic floatation test depends upon:

- a) Change in mass
- b) Change in weight
- c) Change in specific gravity
- d) Change in size

11) Blunt trauma may produce an incised wound when it occurs over which of the following sites:

- a) Scalp b) Hand c) Neck d) Chest

12) In homicidal strangulation by umbilical cord, we can't find:

- a) Abrasions b) Uniform Wharton's jelly
c) Twice twisting around the neck d) Deep constriction marks

13) Gutter wound is:

- a) deep open track uniting the inlet and the exit in subcutaneous tissue

14) Fracture in distal phalange of little finger considered:

- a) disfigurement b) simple injury
c) dangerous injury d) fatal injury

Toxicology

15) Initial acid base disturbance in salicylate toxicity:

- a) metabolic acidosis b) respiratory alkalosis c) metabolic alkalosis d) respiratory acidosis

16) A 35-year-old male was brought to the ER in a comatose state with depressed respiration. His blood alcohol concentration is 500 mg/dL. What is the most important management?

- a) Correction of hypoglycemia, thiamine, and hemodialysis
b) Sodium bicarbonate and methyl pyrazole
c) Gastric lavage to prevent more absorption from GIT

17) A 3-year-old child was brought to the ER after drinking an unknown liquid found under the kitchen sink. He has a characteristic odor, cough, and shallow breathing. Chest X-ray shows bilateral infiltrates. What is the most important initial management?

- a) Immediate gastric lavage must be done
b) Perform gastric lavage with cuffed endotracheal tube
c) Induce emesis immediately to remove the poison from the stomach

18) What is wrong about botulism:

- a) Cranial nerves affection b) Descending symmetrical paralysis
c) No sensory loss d) Disturbances of consciousness

19) Chelators are systemic antidotes of metals by:

- a) Precipitating the metal in the stomach b) Counteract their toxic effect

c) Combine with metal ion forming easily excreted

d) Oxidation of the metal ion to non-toxic complex

20) Atropine causes urine retention and constipation by which of the following mechanism:

a) renal failure

b) CNS depression

c) relaxation of smooth muscle

d) dehydration

21) Target organ of acetaminophen:

a) liver

b) lung

c) heart

d) kidney

22) Skin manifestation of barbiturates:

a) tensed bullae

b) purpura

c) petechia

d) urticaria

23) A man came to the hospital comatosed from CO poisoning and his carboxyhemoglobin level was 50%. The most important treatment is:

a) Hyperbaric oxygen

b) Nitrites

c) Mechanical ventilation

d) 100% oxygen

24) Which of the following has remote and local action ?

a) mercuric chloride

b) sulphuric acid

c) Ammonium hydroxide

d) silica

Written Forensic

1) Thermal hematoma is (1 mark)

2) Absence of current marks in and (1 mark)

3) Mention types of fissure fracture. (2 marks)

4) How to identify the suspected weapon from examination of the missile extracted from the body? (2 marks)

5) Mention types of hymen for medico-legal importance. (2 marks)

6) What's the definition of ecchymosis? and mention two examples. (2 marks)

7) A 19-year-old female was found lying dead at home with a large cut wound across the neck accompanied by multiple small superficial cuts around it. On forensic examination, the body was warm and stiff.

- A) How to verify the age in this case? (1 mark)
 B) Mode of injury? Why? and Cause of death? (2 marks)
 C) Diagnose the postmortem interval. (1 mark)

Written Toxicology

- 1) Carbollurea is (1 mark)
 2) Mention two causes of death in cocaine toxicity. (1 mark)
 3) Mention four major classes of illicit drugs (with one example for each group). (2 marks)
 4) Mention indications of MDAC. (2 marks)
 5) Mention clinical manifestations of mercurial erethism. (2 marks)
 6) Mention lines of management in organophosphorus poisoning. (2 marks)

ANSWERS

1	2	3	4	5	6	7	8	9	10
B	C	C	D	C	A	B	A	A	C
11	12	13	14	15	16	17	18	19	20
A	B	A	C	B	A	B	D	C	C
21	22	23	24						
A	A	A	A						

Written Forensic

- 1) Thermal hematoma is

Extradural collection of blood due to severe burning of the head and characterized by soft, friable, brownish in color, small in size and contains CoHb level.

- 2) Absence of current marks in and

- **large body area.**
- **If the time of contact is only few seconds.**
- **volt is very low.**
- **Arc current**

3) Types of fissure fracture.

- **Linear (diastatic and step like)**
 - **Star shaped or mosaic**
 - **Polar**
 - **Bipolar**
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4) How to identify the suspected weapon from examination of the missile extracted from the body?

- **Bullet and Grooves → Rifled weapon.**
 - **Shots and wad → Non-rifled weapon.**
 - **Primary rifling marks (preliminary marks):** the test and the crime bullets have an identical primary rifling marks (number, depth, breadth and direction of the riflings)
 - **Secondary rifling marks (confirmatory marks):** microscopic scratches found on the surface of the fired bullet (fingerprints for the weapon)
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5) Types of hymen for medico-legal importance.

- Elastic annular hymen.
 - Dentate hymen.
 - Fimbriate hymen.
 - Imperforate hymen.
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6) Definition of ecbolics? and two examples.

Specific abortifacient drugs with selective action on the uterine muscles causing uterine contractions.

Examples:

- Prostaglandins and prostaglandin analogues.
 - Progesterone antagonists (Mifepristone).
 - Ergot (Ergometrine).
 - Quinine (large doses).
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7) Case

A) How to verify the age?

- **X-ray of the lower end of femur (knee).**
- **X-ray of the ischial tuberosity.**

B) Mode of injury? Why? Cause of death?

Mode of injury: **Suicidal.**

Why: **Presence of multiple small superficial cuts (hesitation cuts)** around the main cut wound.

Cause of death: **Hemorrhagic shock.**

C) Diagnose the postmortem interval.

3–8 hours.

Written Toxicology

1) Carboluria is

Greenish-black discoloration of urine due to phenol poisoning.

2) **Causes of death in cocaine toxicity.**

- Cardiac arrhythmias.
 - Acute myocardial infarction.
 - Intracranial hemorrhage.
 - Respiratory failure.
 - Hyperthermia.
 - Status epilepticus.
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3) **Mention four major classes of illicit drugs (with one example for each group).**

1. **CNS stimulants:** Cocaine, Amphetamine, Methamphetamine.
 2. **CNS depressants:** Alcohols, Benzodiazepines, Barbiturates.
 3. **Hallucinogens:** Cannabis, LSD, Khat.
 4. **Narcotics:** Heroin, Morphine, Codeine and other opioids.
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4) **Indications of MDAC.**

- If the toxin has a **long half-life**
 - If the toxin has a significant **enterohepatic circulation** (Digoxin, Phenobarbital, Theophylline)
 - If continuous release of toxin occurs from a **sustained release preparation**
 - If a toxin forms a mass in the gut which is a source of continuous release of toxin as aspirin
 - If the ingestion is too massive to be effectively adsorbed by a single dose of charcoal
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5) **Clinical manifestations of mercurial erethism.**

- Shyness.
 - Vague fear.
 - Unable to work under supervision.
 - Loss of confidence.
 - Loss of power of concentration.
 - Loss of memory.
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6) **Lines of management in organophosphorus poisoning.**

- Emergency ABCD
- Decontamination (gastric lavage)
- Specific antidotes (atropine and oximes)
- Supportive treatment